

1. Introduction

The OLTP system is designed for daily operations such as rentals, payments, and customer management. However, it is not suitable for analytical queries.

A Data Warehouse is designed to support decision-making by organizing data into facts and dimensions, enabling efficient analysis of trends, performance, and behavior.

2. Business Questions

The data warehouse is designed to answer the following analytical questions:

1. Which films are rented most frequently?
2. Which customers rent the most films?
3. Which films are returned late most often?
4. What is the average rental duration per film?
5. Which stores have the highest rental activity?
6. How does rental activity change over time?
7. Which customers generate the highest revenue?
8. Which films generate the highest revenue?
9. How does revenue change over time?
10. Which stores generate the highest revenue?
11. What is the inventory level per store?
12. Which films have the highest stock availability?

These questions help management understand performance, customer behavior, and business trends.

3. Dimensional Model Design

3.1 Business Processes

The main business processes are:

- Rental transactions
- Payment transactions
- Inventory management

3.2 Fact Tables

1. Fact_Rental

- Grain: One record per rental transaction
- Measures:
 - actual_duration
 - is_late
- Foreign Keys:
 - customer_key
 - film_key
 - store_key
 - staff_key
 - date_key

2. Fact_Payment

- Grain: One record per payment
- Measures:
 - payment_amount
- Foreign Keys:
 - customer_key
 - film_key
 - store_key
 - staff_key
 - date_key

3. Fact_Inventory

- Grain: Number of films per store
- Measures:
 - stock_count
- Foreign Keys:
 - film_key
 - store_key

3.3 Dimension Tables

Dim_Customer

- Attributes: full_name
- Source: customer

Dim_Film

- Attributes: title, rental_duration
- Source: film

Dim_Store

- Attributes: store_id
- Source: store

Dim_Staff

- Attributes: full_name
- Source: staff

Dim_Date

- Attributes: year, month
- Source: rental, payment

Schema Type

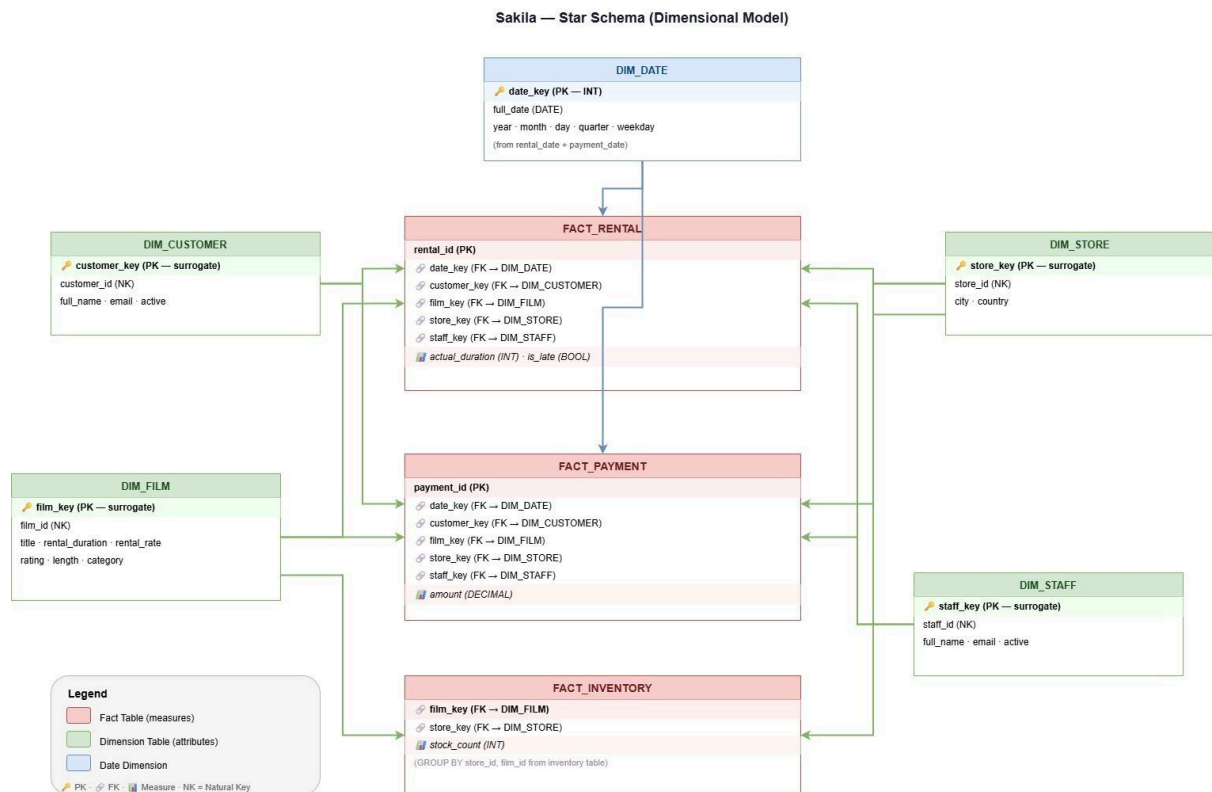
The design follows a Star Schema, where fact tables are connected directly to dimension tables.

4. Dimensional Model Diagram

The model consists of:

- 3 fact tables in the center
- Shared dimensions (Customer, Film, Store, Staff, Date)

Each fact table connects to dimensions using surrogate keys.



5. ETL Design

5.1 Extract

Data is extracted from OLTP tables:

- rental
- payment
- customer
- film
- inventory
- store
- staff

These tables provide all required transactional and descriptive data.

5.2 Transform

Key transformations include:

- Creating `full_name` from first and last names

- Generating surrogate keys
- Creating date_key from dates
- Calculating actual rental duration
- Detecting late returns
- Handling null return dates
- Aggregating inventory data

5.3 Load

- Dimension tables are loaded first
- Fact tables are loaded after linking to dimension keys
- Data is cleaned before loading
- Relationships are maintained using surrogate keys

6. Data Quality Considerations

- Handle missing return dates
- Remove duplicate records
- Ensure valid foreign key relationships
- Standardize data formats
- Validate calculated fields (duration, revenue)

7. Conclusion

The proposed data warehouse improves analytical capabilities by organizing data into facts and dimensions. It supports business decision-making by enabling efficient analysis of rentals, revenue, and inventory.